LUNARSAFE // OFFICIAL MEMORANDUM & SUBMISSION TRANSMITTAL

TEAM ROCKET · AEROX CLUB · PARUL UNIVERSITY, VADODARA

SPACE APPLICATIONS CENTRE (SAC) · INDIAN SPACE RESEARCH ORGANISATION

DATE:	October 3, 2026
TO:	The Director, Space Applications Centre (SAC) Indian Space Research Organisation (ISRO), Ahmedabad - 380015, Gujarat, India
FROM:	Team Rocket · AEROX Club · Parul University, P.O. Limda, Vadodara - 391760
SUBJECT:	**Submission of LunarSafe: Autonomous Lunar Landing-Site Hazard Assessment Platform**

Respected Director,

On behalf of Team Rocket, the student aerospace and robotics division of the AEROX Club at Parul University, Vadodara, under the technical supervision of Project Head Darshan Patil, we have the distinct honour of submitting the complete verification dossiers, scientific methodology package, and source distribution for LunarSafe—an autonomous, deterministic lunar landing-site hazard assessment engine engineered specifically for robotic lunar landers.

1. Project Overview & Motivation

The terminal descent phase of lunar soft landing presents severe autonomous navigational challenges, as demonstrated during the Chandrayaan-2 Vikram descent where terminal guidance limitations and unexpected surface slopes impaired the landing trajectory. While Chandrayaan-3 achieved historic success at Shiv Shakti Point with an enlarged 4.0 km × 2.4 km target ellipse and the upgraded Lander Hazard Detection & Avoidance Camera (LHDAC), future lunar south pole missions targeting permanently shadowed regions (PSRs) require automated, real-time, deterministic hazard classification across extreme topographical gradients. LunarSafe provides an end-to-end deterministic hazard mapping pipeline that eliminates stochastic variances and guarantees identical hazard classifications across heterogeneous mission compute environments.

2. Technological Implementation & Processing Pipeline

We have architected a mathematically rigorous 5-step pipelined state machine that ingests raw planetary elevation data and optical orbital scans to produce flight-grade landing zone decisions:

- Stage 1 (Chromatic Ingestion): Strict panchromatic grayscale verification with automated color-variance rejection.
- Stage 2 (Topographic Derivation): Analytical Horn (1981) 3rd-order weighted 3×3 gradient kernel computing true surface slope.
- Stage 3 (Morphological Hazard Extraction): Multi-scale Sobel spatial operator detecting slope breaklines, crater rims, and boulders.
- Stage 4 (ISRO 24m Standard Raster Grid): Discretization into cell sizes compliant with Chandrayaan-3 landing leg track geometry.
- Stage 5 (Cryptographic Proof of Determinism): SHA-256 HMAC cascading digest chain sealing every stage with Ed25519 digital signatures.

3. Primary Orbital Datasets Utilized

The system was developed and validated directly against authentic planetary observation archives:
• ISRO Chandrayaan-2 TMC-2 Calibrated Swath: High-latitude lunar observation (ch2_tmc_ncn_20260823T1657513854, Orbit 31200) comprising 49,119 lines × 4,000 samples (392,952,000 bytes, 16-bit UnsignedLSB2, 6.40 m/pixel ground sample distance).
• NASA LOLA DEM Ground-Truth Rasters: IEEE 754 Float32 elevation rasters across historic and polar landing sites.

4. Empirical Validation Findings & Benchmark Metrics

Multi-Site LOLA Topographic Validation:	MAE = 0.610° | RMSE = 0.843° across 4 lunar benchmark sites (ISRO threshold < 2.0°: PASS)
Apollo 11 Tranquility Base Slope:	Computed: 1.22° vs Published: 1.30° (Rosenburg et al. 2011) | Error: 0.08°
Shiv Shakti Point (Chandrayaan-3):	Computed: 2.33° vs Published: 2.00° (Durga Prasad et al. 2023) | Error: 0.33°
Morphological Hazard Detection Accuracy:	Precision: 78.33% | Recall: 85.45% | F1-Score: 81.74% (100+ Ground-Truth Features)
Full 392 MB ISRO TMC-2 Swath Analysis:	8,047.66 km² assessed | Safe Area: 82.96% (6,623.81 km²) | 681,937 craters | 244,456 boulders
Primary Touchdown Zone Identified:	6,583.84 km² contiguous zone centered at 73.3414°N, 12.7655°E (ISRO Grade B)
Frozen Engine Verification (Deterministic):	Size: 41,930 bytes | SHA-256: 8CA34C558C41F0B324A207A1D758476D397319999313D59A5919E759128

5. Submission Package Contents (153 Total Files)

Accompanying this memorandum is the comprehensive archive (LunarSafe_ISRO_Submission.zip, 183.81 MB):
1. Validation Reports (01_Validation_Reports): Multi-Site LOLA Benchmark, Hazard Detection, and Independent Audit Reports.
2. Scientific Documentation (02_Scientific_Documentation): 6 publication-quality monographs (Methodology, Algorithms, Validation Results, Error Analysis, Technical Limitations, and Academic References).
3. Full ISRO Swath Analysis (03_Full_ISRO_Analysis): 768-tile Swath Report and calibrated 1000 × 2000 px Hazard Coverage Map.
4. Master Engineering Dossiers (04_Master_Dossiers): Complete Technical and Project Architecture Dossiers.
5. Structured Scientific Datasets (05_Data_and_Maps): Full analysis JSON (768 tiles), CSV metrics, and labeled ground-truth sets.
6. Flight Software Source Code (06_Source_Code): Self-contained TypeScript engine, test suites, and PDS4 raw parser modules.
7. High-Definition Verification Artifacts (07_Screenshots): 25 photographic and terminal evidence records.

6. Request for Technical Review and Scientific Feedback

We respectfully request the esteemed scientists, mission planners, and technical teams at the Space Applications Centre (SAC/ISRO) to review this software platform, validate our algorithmic derivations, and provide feedback on potential adaptations for forthcoming lunar and planetary exploration missions. Team Rocket remains available to demonstrate the platform and assist with any integration inquiries.

Yours respectfully,

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